



POWER_INPUT_14S

J1: 14S auxiliary input from separate protected PDB

TVS + reverse polarity + inrush limiting + input fuse

Voltage/current monitor to autopilot and AI logger

BMS IS EXTERNAL AND IS NOT PART OF THIS PCB

DATA_REQUIRED: max auxiliary current, connector, surge profile

Sheet: /POWER_INPUT_14S/
File: power_input_14s.kicad_sch

Title:

Size: A4	Date:
----------	-------

 KiCad E.D.A. 10.0.6	 E.D.A.
---	--

Date:

Rev:

Id: 2/11

CAN_AND_SENSORS

Two protected CAN-FD branches with selectable termination

Pressure, flow, tank level, temperature and obstacle interfaces

Per-branch power switching and fault telemetry

Grounding and cable shield termination defined at one point

DATA_REQUIRED: sensor list, cable lengths and bus bit rate

Sheet: /CAN_AND_SENSORS/
File: can_and_sensors.kicad_sch

Title:

Size: A4	Date:
----------	-------

Date:

Rev:

Id: 3/11

LOAD_SWITCHES_12V

Ключи 12 V: клапаны форсунок, навигационные огни и anti-collision, вентилятор, зуммер

Перенос схемы бывшего модуля ключей 12 V (источник DATA_REQUIRED)

Обратные диоды / TVS на каждой индуктивной нагрузке, контроль тока ключей

Возвратные токи клапанов не проходят под аналоговой частью CAN_AND_SENSORS

DATA_REQUIRED: число клапанов форсунок и их ток

Sheet: /LOAD_SWITCHES_12V/ File: load_switches_12v.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6	Id: 4/11	

POWER_CONVERSION

Isolated/non-isolated pre-regulator: architecture selection pending

12 V rail: cameras, radio and auxiliary payload

5 V rail: autopilot peripherals and interfaces

AI rail: dedicated protected output, swappable compute module

DATA_REQUIRED: rail currents, ripple, hold-up and thermal limits

Sheet: /POWER_CONVERSION/
File: power_conversion.kicad_sch

Title:

Size: A4	Date:
----------	-------

Rev:

Id: 5/11

AI_COMPUTE_INTERFACE

Swappable companion computer; no direct motor authority

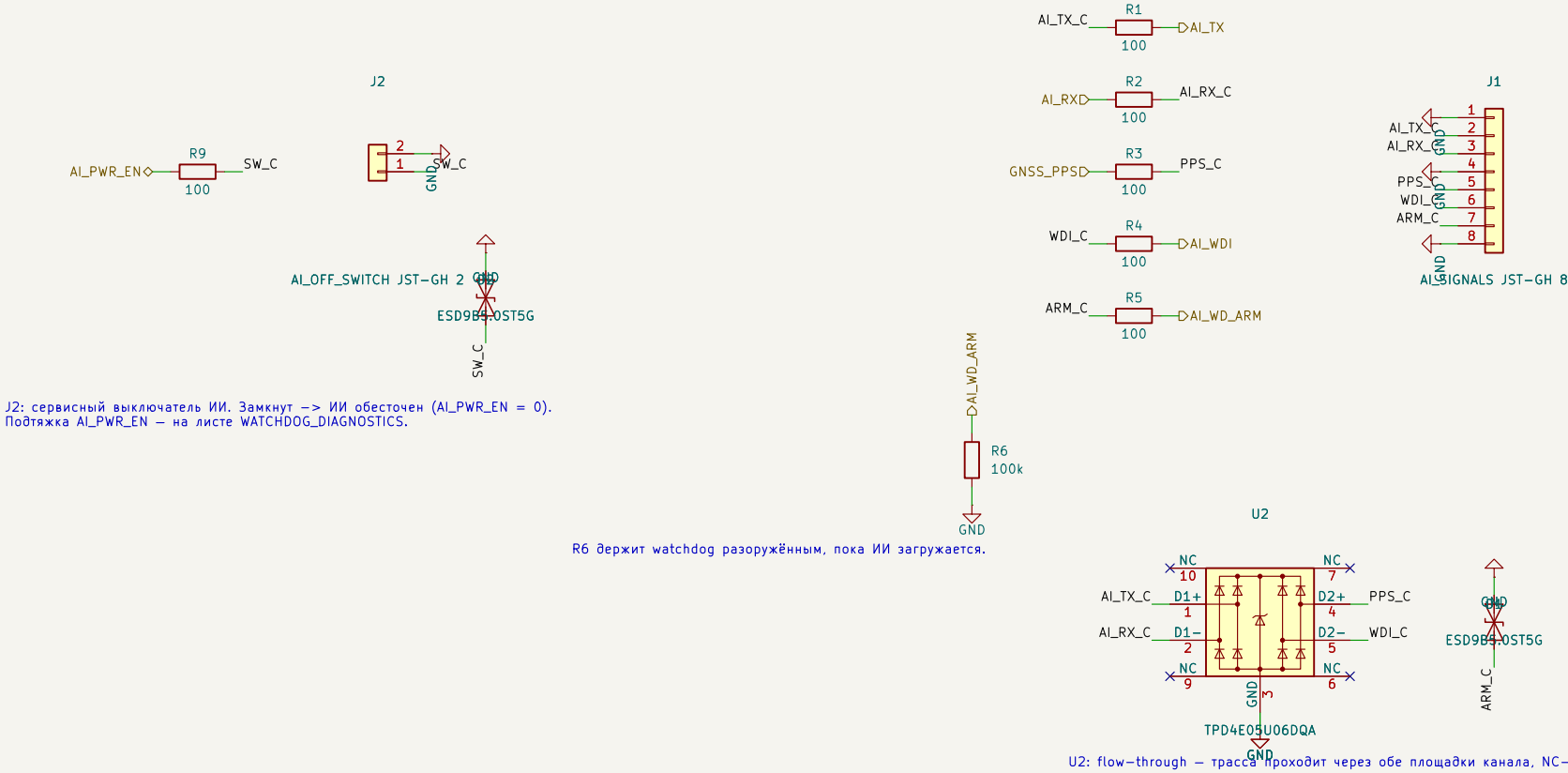
Ethernet/USB/UART/CAN link to flight stack and cameras

Dedicated load switch, current monitor and hardware watchdog

Storage/service connector and controlled shutdown signal

DATA_REQUIRED: AI module, peak power and thermal solution

РАЗЪЁМ ИИ-МОДУЛЯ (вариант А: Jetson Orin NX / вариант Б: Orin Nano, RK3588), уровни 3.3 V.
J1: 1 GND, 2 AI_TX (ИИ -> Pixhawk TELEM2), 3 AI_RX, 4 GND, 5 PPS (GNSS -> ИИ), 6 WDI (импульсы ИИ),
7 WD_ARM (ИИ взводит watchdog после загрузки), 8 GND. Питание ИИ — лист POWER_CONVERSION.



Значения с пометкой CHECK_DATASHEET сверить с документацией
NOT_FOR_MANUFACTURE / NOT_FOR_FLIGHT

АГД6.07.010

Sheet: /AI_COMPUTE_INTERFACE/
File: ai_compute_interface.kicad_sch

Title: AI_COMPUTE_INTERFACE

Size: A3 Date: 2026-09-27

Rev: 0.1 DRAFT

KiCad E.D.A. 10.0.6

Id: 6/11

ETHERNET

Коммутатор 5 портов: Pixhawk 6X, SIYI MK15 air unit, SIYI A2 mini, ИИ–модуль, сервис

Трансформаторная развязка / магнетики на каждом порту, ESD на разъёмах

Диф. пары 100 Ом, вдали от DC/DC листа POWER_CONVERSION

DATA_REQUIRED: микросхема коммутатора (в библиотеке KiCad 5–портового нет) и разъёмы

Sheet: /ETHERNET/ File: ethernet.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.6	Id: 7/11	

FLIGHT_CONTROL_IO

Autopilot remains independent of companion AI computer

Dual CAN, UART telemetry, I2C/SPI service, GPIO and safety input

RC input and emergency stop path bypass AI software

ESD protection and keyed locking connectors

DATA_REQUIRED: exact Pixhawk model and connector pinout

Sheet: /FLIGHT_CONTROL_IO/
File: flight_control_io.kicad_sch

Title:

Size: A4	Date:
----------	-------

Rev:

 KiCad E.D.A. 10.0.6	 E.D.A.
---	--

Id: 8/11

WATCHDOG_DIAGNOSTICS

Hardware watchdog supervises AI power and heartbeat

Independent autopilot rail remains active during AI reset

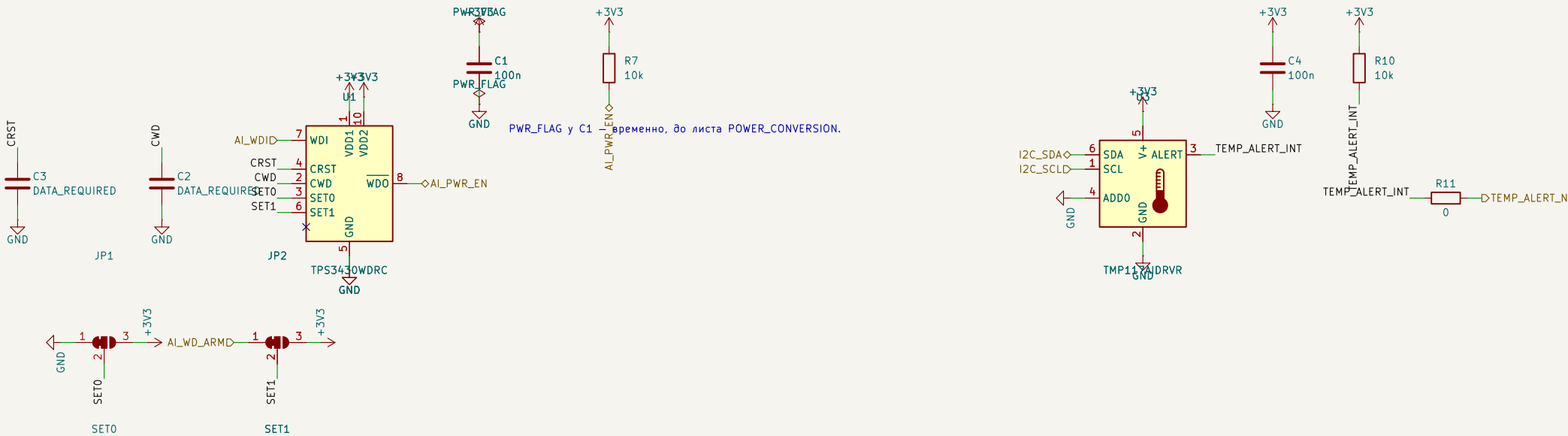
Rail—good, overcurrent and overtemperature telemetry

Service LEDs, test points and black—box event logging

STATUS: DATA_REQUIRED / NOT_FOR_MANUFACTURE / NOT_FOR_FLIGHT

WATCHDOG TPS3430: ИИ шлёт импульсы на WDI. При пропуске -WDO=0 -> AL_PWR_EN=0 -> ключ питания ИИ (лист POWER_CONVERSION) обесточивает ИИ на время tRST (C3), затем включает. Питание Pixhawk не затрагивается. AI_PWR_EN — «монтажное И» (открытый сток): -WDO и сервисный выключатель J2, подтяжка R7. SET0/SET1 — режим окна и включение watchdog по таблице datasheet: CHECK_DATASHEET. JP2 по умолчанию подаёт на SET1 сигнал AI_WD_ARM, JP1 — SET0 на GND.

TMP117: температура отсека электроники / ИИ, I2C 0x48 (ADD0 = GND). Подтяжки I2C — на листе CAN_AND_SENSORS (одна пара на шину).



Значения с пометкой CHECK_DATASHEET сверить с документацией
NOT_FOR_MANUFACTURE / NOT_FOR_FLIGHT

АГД6.07.010

Sheet: /WATCHDOG_DIAGNOSTICS/
File: watchdog_diagnostics.kicad_sch

Title: WATCHDOG_DIAGNOSTICS

Size: A3 Date: 2026-09-27

Rev: 0.1 DRAFT

KiCad E.D.A. 10.0.6

Id: 9/11

RC_AND_TELEMETRY

Primary RC receiver input with independent failsafe path

Telemetry radio and optional LTE/Wi-Fi service modem

GNSS/compass interfaces kept away from DC/DC magnetic fields

External antenna connectors with ESD protection

DATA_REQUIRED: radio region, frequencies and connector selection

Sheet: /RC_AND_TELEMETRY/
File: rc_and_telemetry.kicad_sch

Title:

Size: A4	Date:
----------	-------

Rev:
Id: 10/11

CAMERAS

Front navigation camera and downward observation camera

Optional depth/obstacle camera interface

Protected power switching and synchronization trigger

High-speed links routed away from switching power stages

DATA_REQUIRED: camera models, interface and cable type

Sheet: /CAMERAS/
File: cameras.kicad_sch

Title:

Size: A4	Date:
----------	-------

Rev:
Id: 11/11